VIVIAN LI

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Skills & Interests

Programming & Scripting
Python / R / C++ / Java
SQL / Bash

Libraries & Frameworks
numpy / pandas / matplotlib
scikit-learn / PyTorch / Tensorflow
ggplot

Web Development
Javascript / Node / React / Vue
HTML / CSS / PostgreSQL / PHP

Other
High-performance Computing

Education

University of Colorado Boulder / August 2023 - present
Ph.D. student in Computer Science & Interdisciplinary Quantitative Biology
Advised by Dr. Aaron Clauset (CSCI) & Laura Dee (EBIO)

Cornell University / May 2020
B.A. in Information Science & French, Minor in Computer Science
Cumulative GPA: 3.88

Research Statement

I am interested in how mathematical and computational tools can explain the complexity of biological systems, with a focus on ecosystem dynamics. I use mathematical models, machine learning, and network science to study species interactions and the structure of ecological communities. Furthermore, I develop methods to reduce model uncertainty and translate these models into practical ecosystem management strategies. My research sits at the intersection of ecology and computer science, and I am committed to using computational approaches to tackle pressing ecological questions.

Graduate Research

Predicting Ecosystem Recovery Trajectories / 2023-present

- Simulated ecological disturbances and recovery dynamics using the allometric trophic network model across synthetic and empirical ecological networks
- Analyzed how ecosystem structure and different interaction types shape recovery trajectories
- Characterized how parameter uncertainty propagates through ecological models to affect recovery predictions

Identifying Structural Variants in the Human Genome / 2024-present

- Developed a statistical machine learning method to resolve structural variant calls in short- and long-read sequencing data
- Demonstrated improved classification performance over existing methods on synthetic benchmarking datasets
- Identified hundreds of novel variants across the 1000 Genomes Project samples

Modeling Tau Spread Across a Neuronal Network / 2024

- Created a network-based model of the brain to analyze how structural connectivity influences the spread of misfolded tau proteins in neurodegenerative disease
- Simulated the effects of therapeutic interventions to assess their potential to prevent or mitigate disease progression

Modeling the Evolution of Social Foraging Behavior / 2023

- Built an agent-based model to simulate the evolution of traits linked to social foraging behavior in animal populations

Work Experience

Uncountable / March 2020 - March 2022
Full Stack Engineer

- Improved the Uncountable platform through additions and modifications to all pages across the platform, responding to customer asks and improving the code quality and web design of the platform, in React
- Designed numerous API endpoints to add complex functionalities and improve AI tools for user interactions across the platform, in Python and PostgreSQL
- Designed and implemented an inventory management system and a series of visualizations for our AI tools, and managed the platform's project notebook through UI design and code review

NASA Jet Propulsion Laboratory / June 2019 - August 2019
Frontend & Software Developer Intern, Mars 2020 Mission

- Developed a frontend application to simulate the drive and camera views of the Mars 2020 Rover in a 3D animation upon user input of rover commands, in React
- Simplified the process for users to plan, execute, and verify command sequences in uplink by implementing a live text editor, three.js 3D simulator, and data table into the application
- Worked with various APIs and AWS to run backend computations and deploy the application, and implemented and modified node packages such as react-redux and react-ace

Publications and Presentations

Li, V., Chowdhury, M., Krol, J., Townsend, H., Layer, R. M., & Clauset, A. SPLIT: A Statistical Framework for Resolving Merged Structural Variants. Manuscript in preparation.

Li, V., Dee, L., Clauset, A. Navigating Uncertainties in Modeling Non-Trophic Interaction Dynamics. Presented at ESA Annual Meeting, August 2025.

Li, V., Pablo, L., Li, H., Dee, L., Clauset, A. Quantifying and Eliminating Sources of Uncertainty in Ecological Dynamics Models. Presented at Dynamics Days, January 2026.